Team Mini Project: Working with Streamlit
DS450-01
Data Science Senior Capstone

Project Goal: To write a small web-based application with Pandas, Sci-Kit Learn and Streamlit for viewing/analyzing the Wine Quality data set from the UCI Machine Learning Repository (https://archive.ics.uci.edu/dataset/186/wine+quality)

Scenario: You have been asked by your company to develop a proof of concept for a data set viewing application so your data analysis group can package data for end users to view. Your supervisor suggests that you use Streamlit (https://docs.streamlit.io/) which is a library that supports developing simple web applications in Python.

Instructions: Before working on this project, you must go to an Anaconda Prompt and type: `pip install streamlit` to install Streamlit.

Your application will have several tabs. The information/functionality you need to provide in each tab is described below.

MetaData Tab

This tab should list the meta data of the Wine Quality data as show in .

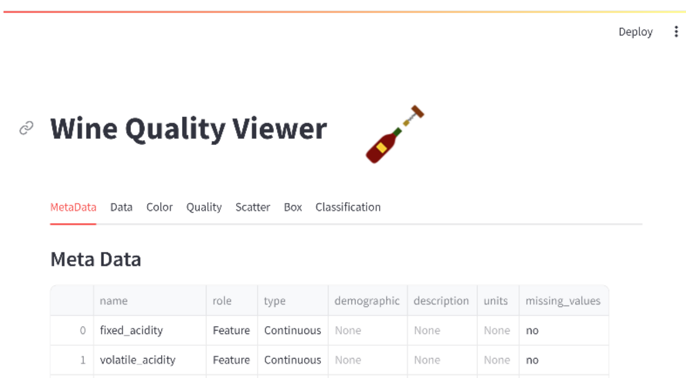

Figure 1: MetaData Tab for Wine Quality Viewer

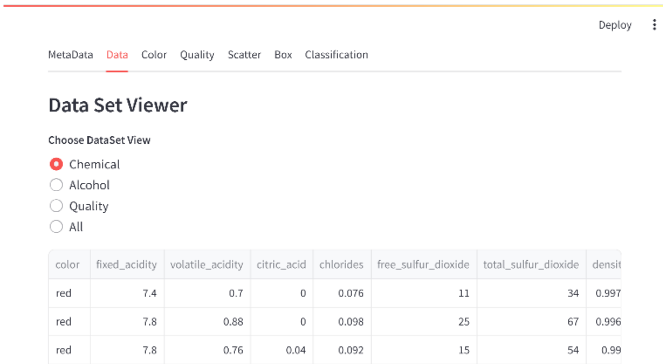

Figure 2: Data Tab for Wine Quality Viewer

Data Tab

This tab should display the columns specified by the radio button according to the following rules:

- Chemical: Should only display color, and then any chemical readings.
- Alcohol: Should only display color and the alcohol column
- Quality: Should only display color and the quality column
- All: Should display all columns with the color column first

When the user clicks a different radio button, the view should change.

Color Tab

This tab should display a bar chart that shows the number of wines of each type (red or white).

Figure 3: Color Tab for Wine Quality Viewer

Quality Tab

This tab should display a bar chart that shows the number of wines of each quality (3 -9).

Figure 4: Quality Tab for Wine Quality Viewer

Scatter Tab

This tab should allow the user to select two of the columns and compare them with a scatterplot. By default, no options should be chosen. Once the user picks two options, the table should display the scatterplot.

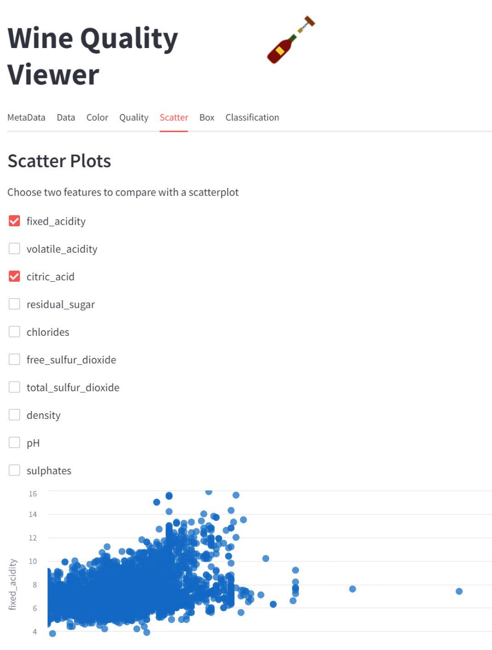

Figure 5: Scatter Tab for Wine Quality Viewer

Box Tab

This table should allow the user to choose which box plot to view. If the user choose a different box plot, it should update the plot.

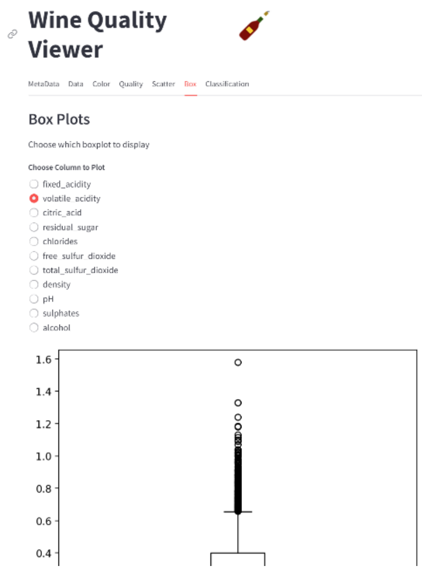

Figure 6: Box Tab for Wine Quality Viewer

Classification Tab

This tab should compute a classification algorithm (wine quality) for the data set and report the results. You may use any classification algorithm you want. Your output should include the classification report and the feature importance.

Figure 7: Classification Tab for Wine Quality Viewer

Submission: Create a new directory in your git repository of this project and put all your project files in that directory. Upload a link to your GitHub repository in the area provided on Moodle by the deadline specified.